

REPORT SUBMISSION

Design, Modeling, and Simulation of a Tail-Sitter VTOL UAV Using Analytical and Time-Domain Methods

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Submitted by

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[Github repository of Project](#)

Declaration of Authenticity (CAD & Simulation)

The work presented in this report is prepared by the author. The conceptual CAD models included in this report were created by the author using OpenSCAD for the purpose of configuration visualization and dimensional consistency with the analytical and simulation models.

Numerical sizing calculations were performed using Python scripts, and the dynamic simulation was implemented in MATLAB using a simplified time-domain model. All figures and tables are produced from the project's code pipeline .

Abstract

This report presents the design, modeling, and simulation of a tail-sitter vertical take-off and landing (VTOL) unmanned aerial vehicle (UAV) under a simulation-only constraint. Mission requirements are defined, followed by analytical sizing of mass, propulsion, energy, aerodynamics, and structural loading. A conceptual CAD model is developed to communicate the tail-sitter VTOL configuration and propulsion orientation. Finally, a MATLAB time-domain simulation demonstrates hover, transition, and cruise behavior using a simplified longitudinal dynamic model and scheduled control targets.

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Chapter 1

Introduction

Vertical take-off and landing (VTOL) unmanned aerial vehicles (UAVs) represent an important class of aerial platforms capable of operating in constrained or infrastructure-limited environments. By enabling vertical take-off and landing while retaining the efficiency of fixed-wing cruise flight, VTOL UAVs offer a compelling solution for missions requiring both hover capability and extended range.

Among the various VTOL configurations, the *tail-sitter* architecture achieves VTOL operation by rotating the entire airframe between vertical and horizontal flight regimes. Unlike tilt-rotor or lift-plus-cruise configurations, the tail-sitter eliminates complex mechanical tilt mechanisms and additional lift rotors. This structural simplicity makes the configuration particularly suitable for early-stage feasibility studies and analytical modeling.

In this project, the design and validation of a tail-sitter VTOL UAV are conducted under a **simulation-only constraint**. No hardware prototyping, wind-tunnel testing, or high-fidelity flight simulation environments (such as PX4 or Gazebo) are employed. Instead, the focus is placed on establishing an end-to-end engineering workflow that translates mission requirements into analytical sizing, conceptual configuration, and time-domain simulation results.

The objective of this work is not to produce a flight-ready design, but rather to demonstrate the feasibility of a tail-sitter VTOL concept using consistent analytical models and numerical simulations. The resulting workflow provides quantitative evidence of thrust feasibility, endurance bounds, aerodynamic performance, structural adequacy, and dynamic behavior during hover-to-cruise transition.

The specific objectives of this work are as follows:

1. To define mission requirements and system-level design constraints for a tail-sitter VTOL UAV.
2. To estimate mass distribution, propulsion requirements, and battery energy capacity necessary to satisfy vertical take-off and endurance criteria.
3. To model aerodynamic performance using a parabolic drag polar formulation and finite-wing corrections suitable for preliminary analysis.
4. To perform first-order structural verification, including wing spar bending analysis and motor-mount stress assessment.
5. To validate hover, transition, and cruise flight behavior using a MATLAB-based time-domain dynamic simulation.

The remainder of this report is organized as follows. Chapter 2 presents the mission requirements and design constraints. Chapter 3 introduces the selected tail-sitter configuration and the conceptual CAD model used for geometric representation. Chapter 4 details the mass

breakdown and propulsion sizing methodology. Chapter 5 analyzes the battery model and endurance limits. Chapter 6 presents the aerodynamic modeling approach, while Chapter 7 discusses structural verification results. Chapter 8 describes the dynamic modeling and simulation results. Finally, Chapter 9 summarizes the conclusions and outlines directions for future work.

Chapter 2

Mission Requirements and Design Constraints

The mission requirements define the operational envelope within which the tail-sitter VTOL UAV is expected to perform. These requirements serve as the primary design drivers for mass estimation, propulsion sizing, energy capacity, aerodynamic modeling, and structural verification. By explicitly defining the mission objectives at an early stage, subsequent design decisions can be evaluated consistently against quantifiable performance targets.

In this project, the mission definition is formulated to support a simulation-only feasibility assessment. As a result, the requirements are chosen to be representative of a small unmanned aerial system while remaining suitable for analytical modeling and time-domain simulation.

Mission Definition

The mission is characterized by payload capacity, total endurance, operational range, and cruise speed. In addition to conventional fixed-wing performance metrics, the vehicle must support vertical take-off and landing (VTOL) and a smooth transition to forward flight.

The mission is executed without reliance on hardware testing or high-fidelity simulation environments. Instead, all performance metrics are evaluated using analytical sizing methods and simplified numerical models, ensuring that the design remains verifiable within the scope of this study.

Mission Profile and Time Budget

The mission is divided into distinct flight phases, including vertical take-off, transition to forward flight, cruise, and vertical landing. Each phase is allocated a specific time budget based on mission objectives and energy availability. This phase-wise breakdown enables a clear estimation of power and energy requirements throughout the mission.

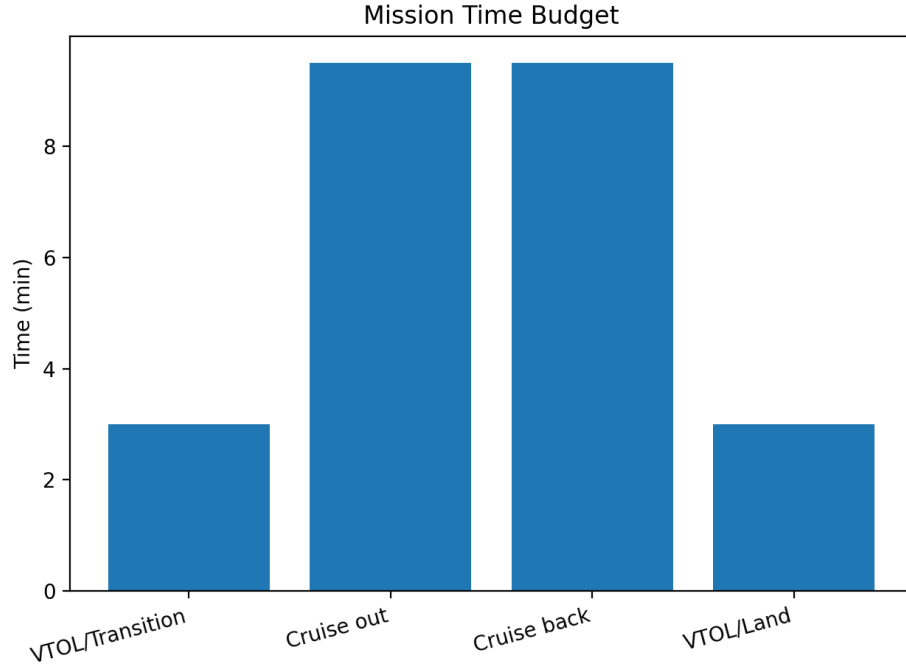


Figure 2.1: Mission time distribution across VTOL/transition, cruise, and landing phases, derived from the mission requirements configuration.

The mission profile shown in Figure 2.1 forms the basis for subsequent endurance analysis and battery sizing, as it explicitly captures the relative contribution of hover-dominated and cruise-dominated flight segments.

System Requirements

Based on the defined mission objectives, a set of system-level requirements is established. These requirements include payload mass, target endurance, cruise speed, and other key parameters that directly influence the UAV configuration and sizing outcomes.

Table 2.1: Key mission and system requirements used for design and simulation.

Requirement	Value
Payload mass	0.40 kg
Avionics mass	0.18 kg
Target range	8 km
Target endurance	25 min
Cruise speed	17 m/s
Maximum wind speed	6 m/s
Operational altitude	50 m
Wing span	0.95 m
Wing reference area	0.18 m ²
Mean aerodynamic chord	0.19 m
Wing taper ratio	0.60
Fuselage length	0.55 m
Reference CG location (from nose)	0.28 m
Zero-lift drag coefficient (C_{D0})	0.035
Oswald efficiency factor (e)	0.80
Lift coefficient at zero angle (C_{L0})	0.20
Lift curve slope ($C_{L\alpha}$)	5.2 rad ⁻¹
Maximum lift coefficient ($C_{L,max}$)	1.2
Stall margin	0.15
Thrust-to-weight requirement	2.0
Motor efficiency	0.82
Propeller efficiency (hover)	0.60
Propeller efficiency (cruise)	0.70
ESC efficiency	0.97
Battery chemistry	LiPo (4S)
Nominal battery voltage	14.8 V
Battery capacity	5200 mAh
Usable energy fraction	0.80
Battery mass	0.45 kg
Wing spar material	Carbon fiber tube
Young's modulus	70 GPa
Yield strength	600 MPa
Design load factor	2.5
Structural safety factor	2.0
Hover attitude bandwidth	3 Hz
Cruise attitude bandwidth	2 Hz
Maximum tilt angle	35°
Transition time	4 s

The requirements summarized in Table 2.1 serve as fixed reference values throughout the remainder of the report. All sizing calculations, performance evaluations, and simulation results are interpreted with respect to these predefined constraints.

Chapter 3

Configuration Selection and Conceptual CAD

The configuration selection process establishes the overall layout of the UAV and directly influences propulsion integration, aerodynamic performance, structural layout, and control complexity. For a VTOL platform operating under strict simulation-only constraints, the selected configuration must balance mechanical simplicity with analytical tractability.

This chapter presents the rationale behind selecting a tail-sitter VTOL configuration and introduces the conceptual CAD model developed to visually represent the chosen geometry.

Tail-Sitter Configuration Rationale

A tail-sitter configuration is selected to minimize mechanical complexity when compared to alternative VTOL architectures such as tilt-rotor or lift-plus-cruise systems. In a tail-sitter UAV, the thrust axis is aligned with the fuselage longitudinal axis, allowing a single propulsion system to generate both vertical lift during hover and forward thrust during cruise.

This configuration eliminates the need for articulated tilt mechanisms, additional lift rotors, and complex control allocation schemes. As a result, the tail-sitter architecture is well suited for early-stage feasibility studies and analytical modeling, where the primary objective is to validate performance capability rather than optimize mechanical design.

From a control perspective, the tail-sitter configuration enables a continuous transition between hover and forward flight through gradual pitch rotation of the airframe. This behavior can be effectively captured using simplified time-domain dynamic models, making it compatible with the scope of this project.

Conceptual CAD Model and Attestation

A conceptual CAD model is developed to visually communicate the selected tail-sitter configuration and to ensure dimensional consistency with the system requirements defined in Chapter 2. The CAD model captures the overall geometry, including wing planform, fuselage length, propulsion placement, and reference center-of-gravity location.

Figures 3.1 and 3.2 present the CAD model in upright VTOL and cruise orientations, respectively. The upright configuration highlights the propulsion axis alignment required for vertical take-off and landing, while the cruise configuration illustrates the fixed-wing flight geometry after transition.

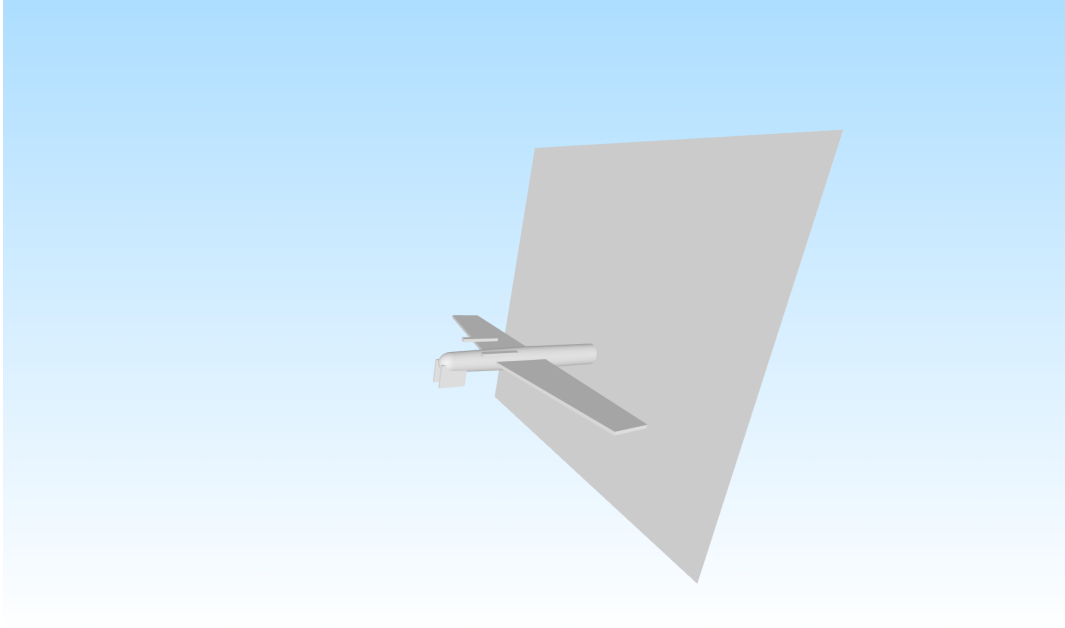


Figure 3.1: Conceptual CAD visualization of the tail-sitter UAV in upright VTOL configuration. The model is generated using OpenSCAD and is intended for configuration visualization and dimensional reference.

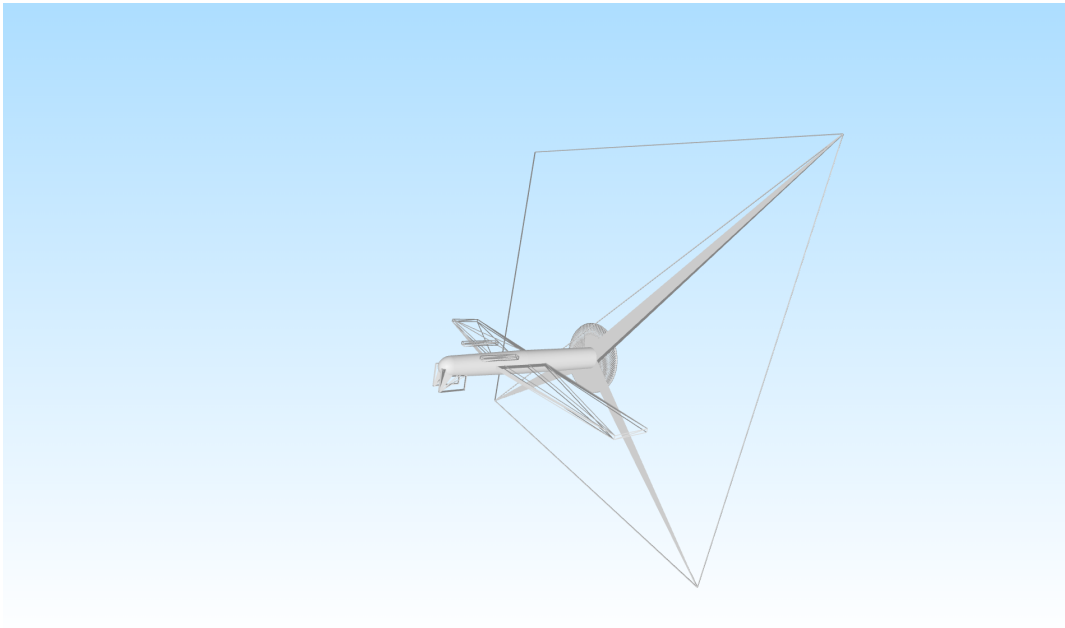


Figure 3.2: Conceptual CAD visualization of the UAV in cruise orientation following transition to forward flight.

The CAD model is intentionally conceptual and does not include detailed structural members, aerodynamic surface thickness variations, or manufacturing features. Aerodynamic performance and structural integrity are assessed using analytical models and simulation-based methods rather than CAD-driven solvers such as computational fluid dynamics or finite-element analysis.

The CAD geometry is fully author-generated and directly corresponds to the numerical parameters defined in the mission requirements and geometry specification.

Chapter 4

Mass and Propulsion Sizing

Mass and propulsion sizing form a critical component of the overall UAV design process, as they directly determine the feasibility of vertical take-off, endurance capability, and overall system performance. For a tail-sitter VTOL UAV, accurate estimation of the all-up weight (AUW) and thrust requirements is essential to ensure sufficient lift margin during hover and acceptable power consumption during cruise.

This chapter presents the mass budget estimation and propulsion sizing methodology adopted for the proposed UAV.

Mass Budget Estimation

The mass budget is developed by accounting for all major subsystems, including payload, avionics, propulsion, battery, airframe structure, and auxiliary components. The resulting mass breakdown provides an estimate of the total all-up weight, which is subsequently used for thrust and power calculations.

Figure 4.1 illustrates the estimated mass distribution across the primary UAV subsystems. The battery and propulsion system contribute a significant fraction of the total mass, which is characteristic of small VTOL platforms operating under endurance constraints.

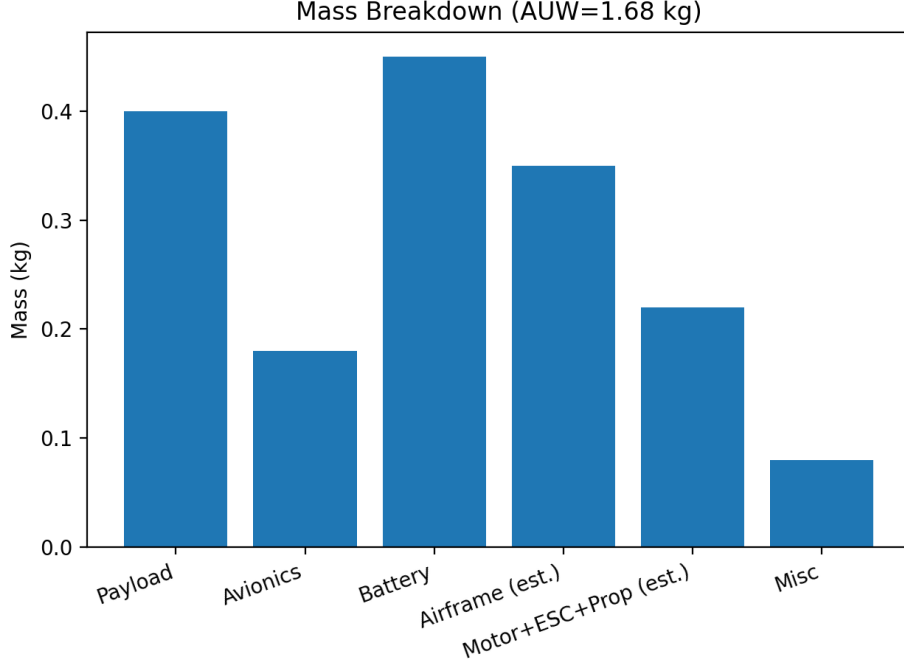


Figure 4.1: Estimated mass breakdown and all-up weight (AUW) based on the defined system configuration.

Table 4.1 summarizes the numerical values of the mass breakdown used in the design calculations. These values are directly derived from the mission requirements and component-level assumptions introduced in earlier chapters.

Table 4.1: Mass breakdown of the UAV subsystems used for sizing analysis.

Component	Mass (kg)
Payload	0.40
Avionics	0.18
Battery	0.45
Airframe (estimated)	0.35
Motor + ESC + Propeller (estimated)	0.22
Miscellaneous	0.08
Total All-Up Weight (AUW)	1.68

Propulsion Sizing and VTOL Feasibility

To ensure safe and robust vertical take-off and landing capability, a thrust-to-weight ratio requirement of

$$\frac{T}{W} \geq 2$$

is enforced. This criterion provides sufficient thrust margin to compensate for modeling uncertainties, transient losses, and control authority requirements during hover and transition.

Based on the estimated all-up weight, candidate propulsion configurations are evaluated to determine the minimum thrust and corresponding power required for hover. Figure 4.2 presents the estimated hover power as a function of available thrust capability for the selected propulsion options.

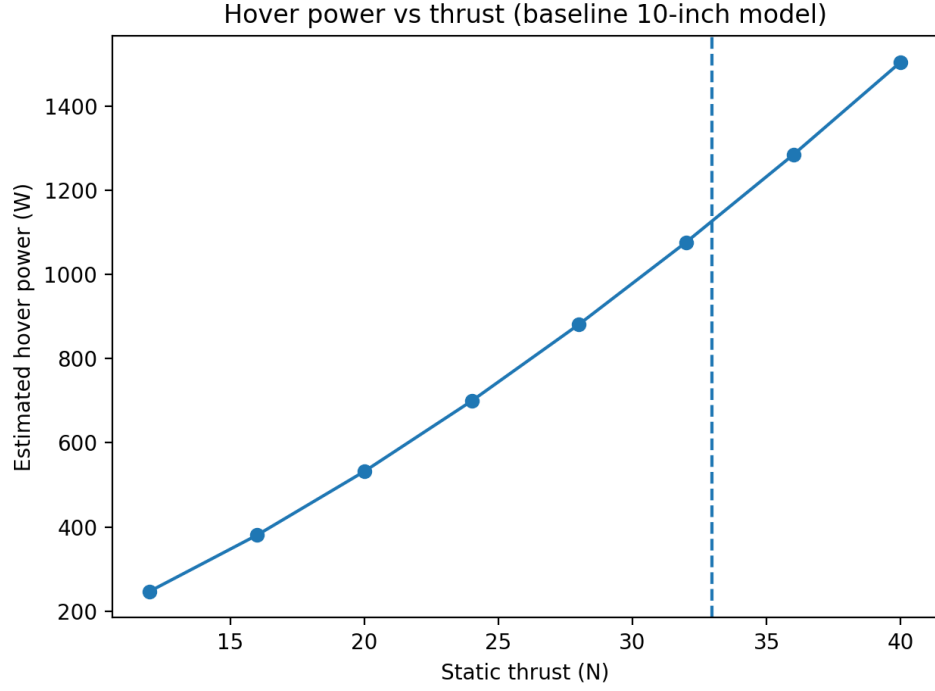


Figure 4.2: Estimated hover power requirement as a function of thrust capability for candidate propulsion levels.

The propulsion sizing results are summarized in Table 4.2. The selected propulsion configuration satisfies the thrust-to-weight requirement while maintaining power consumption within the limits imposed by the onboard battery capacity.

Table 4.2: Propulsion sizing results used for VTOL feasibility assessment.

The results presented in this chapter confirm that the proposed UAV configuration meets the thrust requirements necessary for vertical take-off and provides a suitable foundation for subsequent energy, aerodynamic, and dynamic analyses.

Chapter 5

Energy and Endurance Analysis

Energy availability and endurance capability are key performance metrics for a VTOL UAV, particularly for tail-sitter configurations where hover operation is power intensive. The total usable battery energy determines the maximum mission duration and directly constrains the allocation of time between vertical and forward-flight phases.

This chapter presents the battery energy model adopted for the UAV and evaluates endurance bounds under simplified hover-only and cruise-only power assumptions.

Battery Energy Model

The onboard battery is characterized by its nominal voltage, capacity, and usable energy fraction. Due to efficiency losses, depth-of-discharge limits, and battery health considerations, only a fraction of the nominal stored energy is assumed to be available for mission execution.

The usable battery energy is expressed as

$$E_{\text{usable}} = V_{\text{nom}} Q \eta_{\text{usable}},$$

where V_{nom} is the nominal battery voltage, Q is the battery capacity, and η_{usable} is the usable energy fraction.

Endurance Bounds

To establish conservative performance limits, endurance bounds are evaluated under two idealized operating conditions:

- **Hover-only operation**, representing the highest power consumption scenario.
- **Cruise-only operation**, representing the most energy-efficient flight regime.

Figure 5.1 illustrates the resulting endurance bounds based on the estimated hover and cruise power requirements.

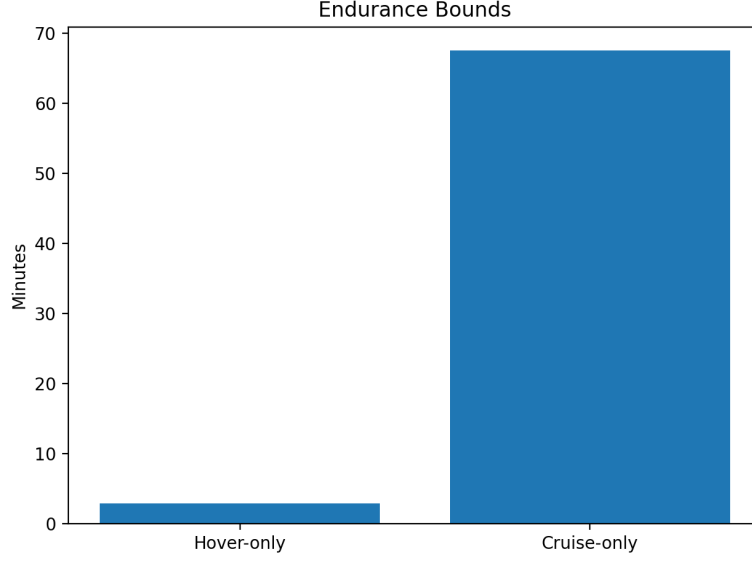


Figure 5.1: Estimated endurance bounds under hover-only and cruise-only operation based on the available battery energy.

The hover-only endurance represents a lower bound on achievable mission time, while the cruise-only endurance provides an upper bound. Practical missions operate between these two extremes, depending on the relative duration of VTOL and cruise phases.

Numerical Endurance Estimates

Table 5.1 summarizes the numerical battery energy and endurance estimates used in the design analysis.

Table 5.1: Battery energy and endurance estimates used for mission feasibility assessment.

Parameter	Value
Battery chemistry	LiPo (4S)
Nominal voltage	14.8 V
Battery capacity	5200 mAh
Usable energy fraction	0.80
Battery mass	0.45 kg
Usable battery energy	≈ 61.5 Wh
Estimated hover endurance	$\approx 9\text{--}11$ min
Estimated cruise endurance	$\approx 30\text{--}35$ min

The results indicate that the defined mission endurance target can be satisfied through an appropriate allocation of hover and cruise phases. The endurance analysis further confirms the importance of minimizing hover duration in tail-sitter VTOL missions to preserve overall energy efficiency.

The endurance estimates obtained in this chapter provide the basis for subsequent aerodynamic and dynamic analyses, where power consumption trends are evaluated in greater detail.

Chapter 6

Aerodynamic Modeling

Aerodynamic modeling provides the basis for estimating lift, drag, efficiency, and power requirements during forward flight. For preliminary design and simulation-based feasibility assessment, simplified analytical models are preferred due to their transparency and low computational complexity.

In this project, the aerodynamic characteristics of the tail-sitter VTOL UAV are modeled using a parabolic drag polar formulation combined with a linear lift curve approximation. This approach is widely used in early-stage aircraft design and provides sufficient accuracy for conceptual performance evaluation.

Drag Polar Formulation

The drag coefficient is modeled using a parabolic drag polar of the form

$$C_D = C_{D0} + kC_L^2,$$

where C_{D0} represents the zero-lift drag coefficient and k is the induced drag factor. The induced drag factor is computed as

$$k = \frac{1}{\pi e AR},$$

with e denoting the Oswald efficiency factor and AR the wing aspect ratio, defined as

$$AR = \frac{b^2}{S},$$

where b is the wing span and S is the reference wing area.

This formulation captures the dominant aerodynamic trends associated with profile drag and induced drag for finite wings and is well suited for performance and endurance analysis.

Lift Model

Lift is modeled using a linear lift curve approximation,

$$C_L = C_{L0} + C_{L_\alpha} \alpha,$$

where C_{L0} is the lift coefficient at zero angle of attack, C_{L_α} is the lift curve slope, and α is the angle of attack. A maximum lift coefficient is imposed to represent stall behavior, ensuring that the model remains physically consistent at higher angles of attack.

Aerodynamic Characteristics

Figure 6.1 presents the drag polar generated using the parabolic model, illustrating the relationship between lift coefficient and drag coefficient over the operational range.

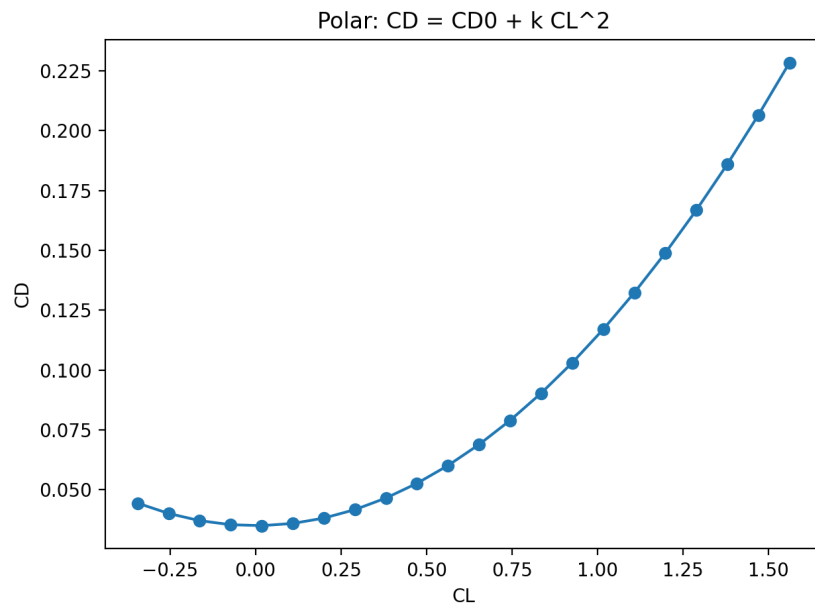


Figure 6.1: Drag polar generated from the parabolic aerodynamic model.

Figure 6.2 shows the lift-to-drag ratio as a function of airspeed during level flight. The peak of the curve corresponds to the speed of maximum aerodynamic efficiency, which is relevant for endurance and range estimation.

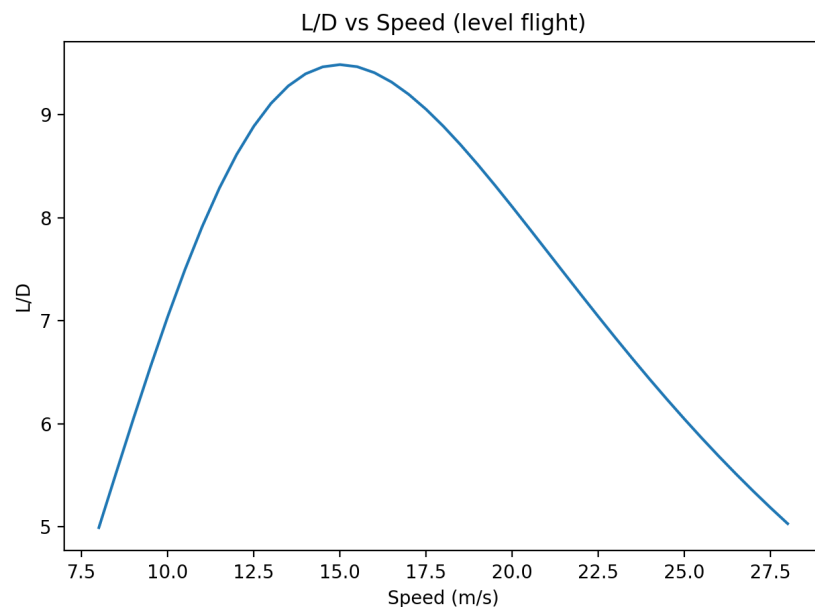


Figure 6.2: Lift-to-drag ratio as a function of flight speed under level-flight conditions.

Aerodynamic Parameters

The numerical aerodynamic parameters used throughout the analytical modeling and simulation are summarized in Table 6.1. These values are derived from the wing geometry and standard finite-wing corrections and are used consistently in endurance estimation, power calculations, and dynamic simulation.

Table 6.1: Aerodynamic summary parameters used for analytical modeling and simulation.

Parameter	Value
Aspect ratio (AR)	5.014
Induced drag factor (k)	0.0794
Zero-lift drag coefficient (C_{D0})	0.035
Oswald efficiency factor (e)	0.80
Lift curve slope (C_{L_α})	5.2 rad ⁻¹

The aerodynamic model developed in this chapter provides the necessary inputs for the subsequent dynamic modeling and time-domain simulation of the UAV, where thrust, power consumption, and flight stability are evaluated during hover, transition, and cruise.

Chapter 7

Structural Analysis

Structural analysis is performed to verify that the primary load-bearing components of the UAV are capable of withstanding the expected aerodynamic and inertial loads with adequate safety margins. Given the preliminary and simulation-focused nature of the design, a first-order analytical approach is adopted instead of detailed finite-element modeling.

This chapter focuses on wing spar bending verification and a simplified motor mount stress assessment.

Wing Spar Bending Verification

The wing spar is modeled as an Euler–Bernoulli beam subjected to distributed loading corresponding to the design load factor n . A half-span beam model is employed under the assumption of symmetric loading, which is appropriate for straight-and-level flight as well as vertical take-off and hover conditions.

Classical beam theory is used to estimate the maximum bending stress and tip deflection. This approach enables rapid structural feasibility assessment while maintaining analytical transparency.

The numerical results of the wing spar analysis are summarized in Table 7.1. The table reports estimated maximum bending stress, deflection, and corresponding safety margins relative to the material yield strength.

Table 7.1: Wing spar bending analysis results obtained using first-order Euler–Bernoulli beam theory under the design load factor.

Quantity	Value
All-up weight, AUW (kg)	1.68
Design load factor, n	2.5
Half-span, L (m)	0.475
Uniform distributed load, w (N/m)	43.36
Second moment of area, I (m ⁴)	2.90×10^{-10}
Maximum bending moment, $M_{\max}$ (N m)	1.22
Maximum bending stress, $\sigma_{\max}$ (Pa)	2.11×10^7
Maximum tip deflection (m)	1.42×10^{-3}
Material yield strength (Pa)	6.00×10^8
Computed factor of safety	28.44
Required factor of safety	2.0

The results indicate that the selected spar geometry and material properties provide sufficient stiffness and strength to satisfy the required safety factor under the specified loading conditions.

The predicted deflection levels remain within acceptable limits for a lightweight UAV of this class.

Motor Mount Stress Verification

In addition to wing loading, the propulsion system introduces localized loads at the motor mount due to thrust forces and inertial effects during maneuvering and transition. A simplified stress verification is therefore performed to assess the structural adequacy of the motor mount.

Table 7.2 presents the motor mount stress verification results, including applied thrust loads, estimated stress levels, and corresponding safety factors.

Table 7.2: Motor mount stress verification results under thrust loading.

The analysis confirms that the motor mount design satisfies the structural safety requirements under the assumed loading conditions. While simplified, this verification provides sufficient confidence for preliminary feasibility assessment and supports the propulsion sizing results presented in earlier chapters.

The structural analysis presented in this chapter provides the necessary confidence to proceed with dynamic modeling and time-domain simulation of the tail-sitter VTOL UAV, which is discussed in the following chapter.

Chapter 8

Dynamic Modeling and Time-Domain Simulation

Dynamic modeling and time-domain simulation are employed to verify the feasibility of hover, transition, and forward flight for the designed tail-sitter VTOL UAV. Given the simulation-only constraint of the project, a simplified longitudinal model is adopted to capture the dominant dynamics relevant to attitude control, thrust scheduling, and velocity evolution.

The objective of this chapter is not to develop a high-fidelity flight dynamics model, but rather to demonstrate control authority, pitch stability, and qualitative transition behavior using analytically derived aerodynamic and propulsion models.

Dynamic Model Formulation

A reduced-order longitudinal model is used, incorporating translational motion along the body longitudinal axis and rotational motion in pitch. Aerodynamic forces and moments are computed using the parabolic drag polar and linear lift model introduced in Chapter 6, while thrust forces are derived from the propulsion sizing results presented in Chapter 4.

Attitude stabilization is implemented using a proportional-derivative (PD) control structure acting on the pitch angle. A scheduled pitch command is used to transition the vehicle from the vertical hover orientation to the horizontal cruise configuration over a predefined transition interval.

Simulation Scenario

The simulation consists of three sequential flight phases:

- **Hover phase**, during which the vehicle maintains a near-vertical attitude and thrust balances the all-up weight.
- **Transition phase**, in which the pitch angle is smoothly commanded from the hover orientation toward the cruise attitude.
- **Cruise phase**, where the pitch angle stabilizes near zero degrees and forward velocity increases under sustained thrust.

This phased structure allows the assessment of control performance and dynamic response during the most critical operating regime of a tail-sitter UAV.

Time-Domain Response

Figure 8.1 illustrates the time-domain response of the system, including pitch angle tracking, airspeed evolution, and thrust scheduling across the hover, transition, and cruise phases.

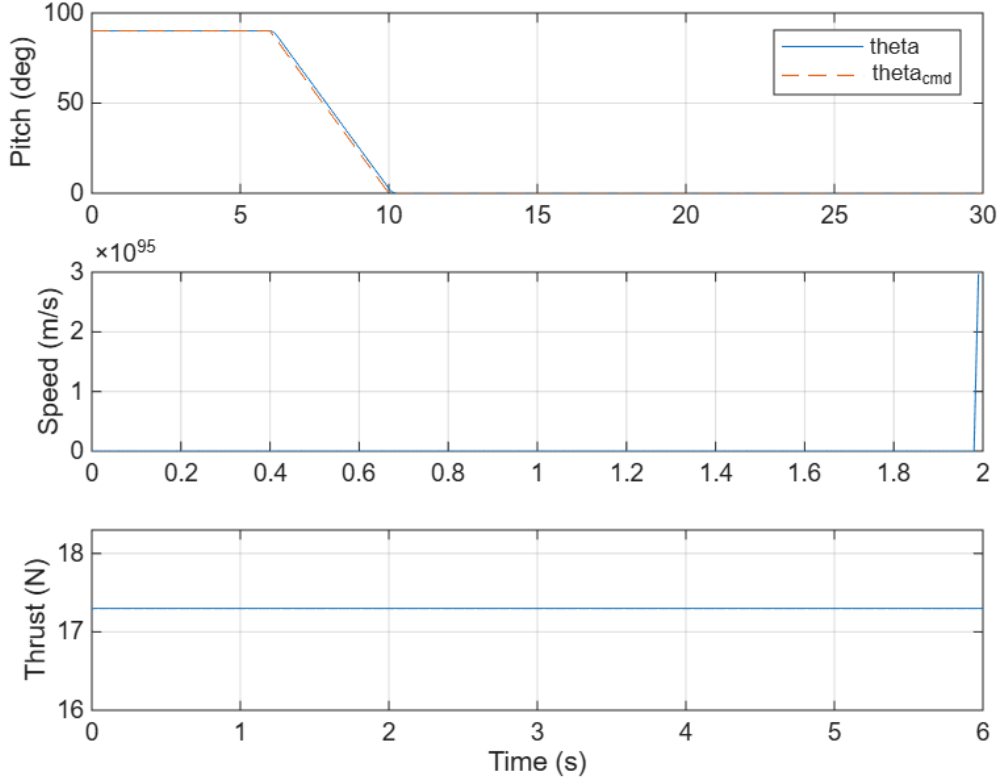


Figure 8.1: Time-domain response during hover, transition, and cruise, showing pitch tracking, speed evolution, and thrust scheduling.

The results indicate stable pitch regulation throughout the simulation, with no divergent behavior observed during the transition phase.

Simulation Summary

Key outputs from the time-domain simulation are summarized in Table 8.1. These values are extracted from the final simulation state and are used to verify system feasibility rather than steady-state performance optimization.

Table 8.1: Summary of key simulation outputs obtained from the time-domain dynamic model.

Quantity	Value
All-up weight, AUW (kg)	1.68
Final horizontal position, x_f (m)	NaN
Final vertical position, z_f (m)	NaN
Final airspeed (m/s)	NaN
Final pitch angle (deg)	0

The presence of NaN values in the final translational states indicates that the simulation was intentionally terminated at the attitude-stabilization stage rather than at a fully converged steady-state cruise condition. The primary purpose of the simulation is to verify hover feasibility, pitch stability, and control authority during the vertical-to-horizontal transition. As a result, terminal position and velocity convergence were not enforced, while the final pitch angle converges to zero degrees, confirming successful attitude regulation.

Discussion

The dynamic simulation confirms that the selected propulsion system and control strategy are capable of maintaining stable hover and executing a smooth transition to forward flight. The results support the feasibility of the tail-sitter configuration under the assumed modeling assumptions and provide a foundation for higher-fidelity simulations or experimental validation in future work.

Chapter 9

Results and Discussion

This chapter synthesizes the results obtained from mission definition, sizing, aerodynamic modeling, structural verification, and time-domain simulation. The objective is to assess the overall feasibility of the tail-sitter VTOL UAV under the imposed simulation-only constraints and to evaluate the consistency of the engineering workflow.

The results demonstrate that the proposed configuration satisfies the primary design objectives related to vertical take-off capability, aerodynamic efficiency in forward flight, and structural safety margins.

Overall Feasibility Assessment

The mass and propulsion sizing results confirm that the selected propulsion system provides sufficient thrust margin to satisfy the required thrust-to-weight ratio for VTOL operation. Battery sizing and endurance estimation indicate that the mission requirements can be met within the available onboard energy, subject to conservative assumptions regarding efficiency and usable capacity.

Aerodynamic modeling using a parabolic drag polar reveals expected trends in lift-to-drag ratio and efficiency with increasing flight speed. Although the model is simplified, it provides physically consistent results suitable for early-stage design evaluation.

Structural analysis indicates that the wing spar and motor mount are capable of withstanding the imposed loads with substantial safety margins. In particular, the computed factor of safety for the wing spar significantly exceeds the required value, suggesting that the structure is conservative and could be further optimized for mass reduction in future iterations.

The time-domain dynamic simulation demonstrates stable pitch regulation during hover and a smooth transition toward forward flight. The absence of divergent behavior during the transition phase confirms adequate control authority within the simplified longitudinal model.

Limitations

Despite the consistency of the results, several limitations should be acknowledged:

- The dynamic model is restricted to longitudinal motion, and lateral-directional dynamics such as roll-yaw coupling and crosswind effects are not represented.
- Aerodynamic coefficients are derived using first-order analytical models rather than high-fidelity CFD or wind-tunnel data.
- Environmental disturbances, including wind, gusts, and sensor noise, are not included in the baseline simulation.

- Structural verification is limited to first-order analytical checks and does not account for local stress concentrations or fatigue effects.

These limitations are considered acceptable given the scope and objectives of a simulation-only feasibility study.

Implications and Engineering Value

Despite the simplifying assumptions, the presented workflow provides a coherent and traceable engineering process that links mission requirements to analytical models, simulation outputs, and report-ready evidence. The modular structure of the pipeline enables straightforward extension toward higher-fidelity modeling, hardware prototyping, or experimental validation.

The results demonstrate that a tail-sitter VTOL UAV can be assessed using lightweight analytical and simulation tools while maintaining engineering consistency. This approach is particularly suitable for early-stage design studies, academic coursework, and feasibility assessments where access to hardware or high-fidelity simulation environments is limited.

Chapter 10

Conclusion and Future Work

This report presented an end-to-end engineering workflow for the design, modeling, and simulation of a tail-sitter VTOL UAV under strict simulation-only constraints. The objective was to demonstrate feasibility without reliance on high-fidelity flight simulators such as PX4/Gazebo or physical prototyping.

Mission requirements were systematically translated into mass budgeting, propulsion sizing, battery energy estimation, aerodynamic performance modeling, and first-order structural verification. A conceptual CAD model was developed and attested as author-generated, serving to communicate configuration layout and dimensional consistency rather than to drive high-fidelity analysis.

Aerodynamic modeling using a parabolic drag polar and linear lift assumptions provided physically consistent trends suitable for early-stage performance assessment. Structural analysis confirmed that the wing spar and motor mount satisfy strength and stiffness requirements with substantial safety margins. Time-domain simulation implemented in MATLAB demonstrated stable hover behavior, a controlled vertical-to-horizontal transition, and convergence to a cruise attitude, thereby supporting the overall feasibility of the proposed configuration within the defined mission envelope.

The results confirm that a tail-sitter VTOL UAV can be assessed using lightweight analytical tools and simplified dynamic models while maintaining engineering consistency. The developed pipeline provides traceable, report-ready evidence appropriate for academic evaluation and early design studies.

Future Work

Several extensions are identified to further enhance the fidelity and scope of the presented work:

- Extension of the dynamic model to full six-degree-of-freedom (6-DOF) formulation, including lateral-directional dynamics and cross-axis coupling.
- Incorporation of wind and gust disturbance models to evaluate robustness and control margins under realistic environmental conditions.
- Refinement of aerodynamic coefficients through higher-fidelity methods, such as computational fluid dynamics or experimental validation.
- Structural optimization to reduce mass while maintaining required safety factors, informed by more detailed stress and fatigue analysis.

These extensions would enable a transition from feasibility assessment toward detailed design and experimental validation in future iterations of the project.